



1. Description

1.1. Project

Project Name	H7R7L8H8
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	04/20/2026

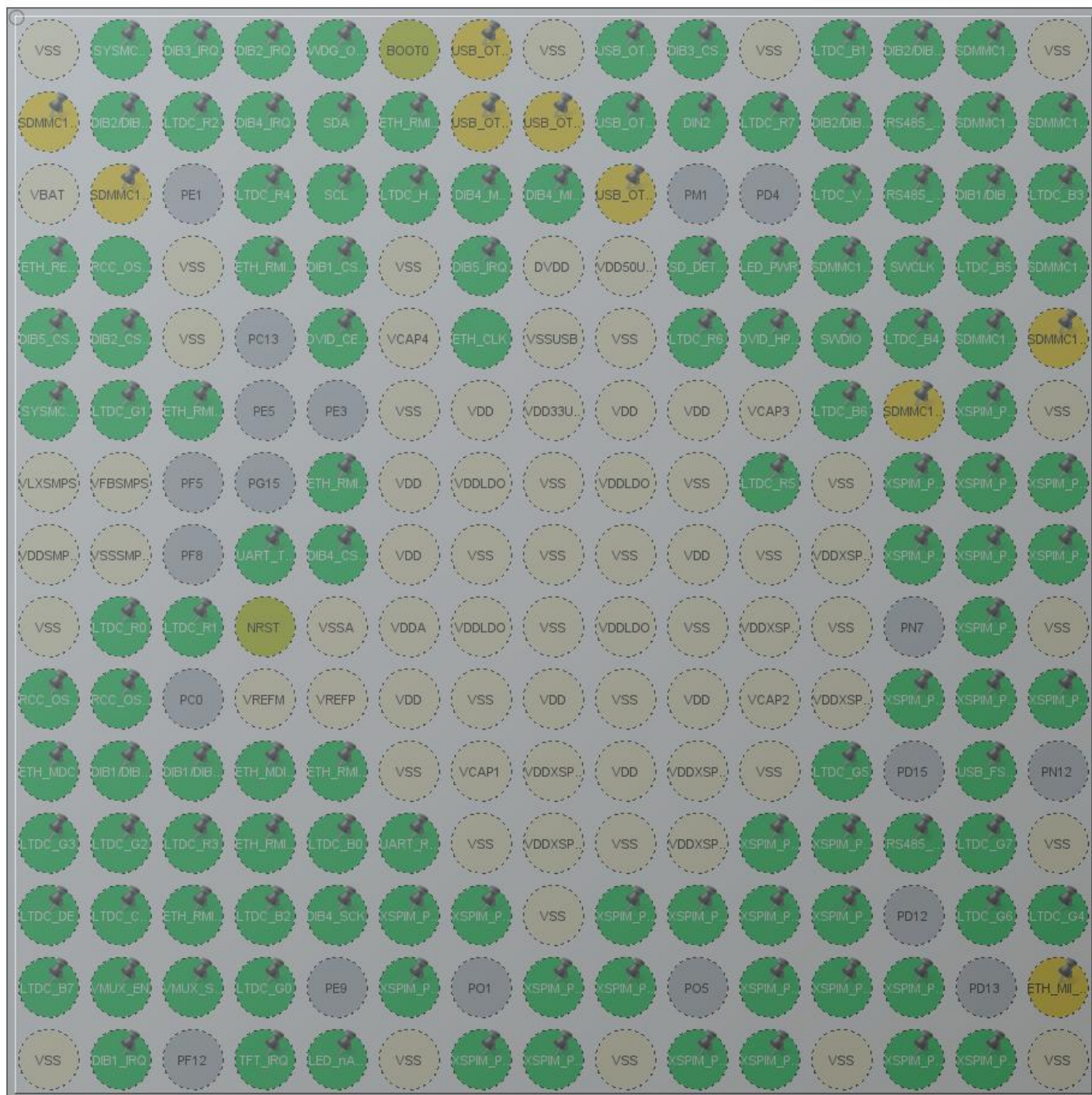
1.2. MCU

MCU Series	STM32H7RS
MCU Line	STM32H7R7/7S7
MCU name	STM32H7R7L8HxH
MCU Package	TFBGA225 HEXA SMPS
MCU Pin number	225

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



TFBGA225 HEXA SMPS (Top view)

3. Pins Configuration

Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A2	PE0	I/O	GPIO_EXTI0	SYSMCU_IRQ
A3	PF3	I/O	GPIO_EXTI3	DIB3_IRQ
A4	PF2	I/O	GPIO_EXTI2	DIB2_IRQ
A5	PG3 *	I/O	GPIO_Output	WDG_OUT
A6	BOOT0	Boot		
A7	PM12 **	I/O	USB_OTG_FS_DM	
A8	VSS	Power		
A9	PM6	I/O	USB_OTG_HS_DP	
A10	PM3 *	I/O	GPIO_Output	DIB3_CSA
A11	VSS	Power		
A12	PD3	I/O	LTDC_B1	
A13	PE14	I/O	SPI4_MOSI	DIB2/DIB5_MOSI
A14	PD2	I/O	SDMMC1_CMD	
A15	VSS	Power		
B1	PB8 **	I/O	SDMMC1_D4	
B2	PE2	I/O	SPI4_SCK	DIB2/DIB5_SCK
B3	PB5	I/O	LTDC_R2	
B4	PF4	I/O	GPIO_EXTI4	DIB4_IRQ
B5	PF0	I/O	I2C2_SDA	SDA
B6	PD7	I/O	ETH_RMII_REF_CLK	
B7	PM11 **	I/O	USB_OTG_FS_DP	
B8	PM9 **	I/O	USB_OTG_HS_ID	
B9	PM5	I/O	USB_OTG_HS_DM	
B10	PM2 *	I/O	GPIO_Input	DIN2
B11	PG0	I/O	LTDC_R7	
B12	PE13	I/O	SPI4_MISO	DIB2/DIB5_MISO
B13	PD1	I/O	UART4_TX	RS485_TX/DOUT3
B14	PC10	I/O	SDMMC1_D2	
B15	PC11	I/O	SDMMC1_D3	
C1	VBAT	Power		
C2	PB9 **	I/O	SDMMC1_D5	
C4	PB3(JTDO/TRACESWO)	I/O	LTDC_R4	
C5	PF1	I/O	I2C2_SCL	SCL
C6	PG2	I/O	LTDC_HSYNC	
C7	PM13	I/O	SPI5_MOSI	DIB4_MOSI

Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
C8	PM14	I/O	SPI5_MISO	DIB4_MISO
C9	PM8 **	I/O	USB_OTG_HS_VBUS	
C12	PE11	I/O	LTDC_VSYNC	
C13	PD0	I/O	UART4_RX	RS485_RX/DIN3
C14	PA12	I/O	SPI2_SCK	DIB1/DIB3_SCK
C15	PA11	I/O	LTDC_B3	
D1	PC15- OSC32_OUT(OSC32_OUT) *	I/O	GPIO_Output	ETH_RESET
D2	PC14- OSC32_IN(OSC32_IN)	I/O	RCC_OSC32_IN	
D3	VSS	Power		
D4	PB7	I/O	ETH_RMII_TXD1	
D5	PB4(NJTRST)	I/O	SPI2_NSS	DIB1_CSA
D6	VSS	Power		
D7	PD5	I/O	GPIO_EXTI5	DIB5_IRQ
D8	DVDD	Power		
D9	VDD50USB	Power		
D10	PM0 *	I/O	GPIO_Input	SD_DETECT
D11	PE15 *	I/O	GPIO_Output	LED_PWR
D12	PC12	I/O	SDMMC1_CK	
D13	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	SWCLK
D14	PA9	I/O	LTDC_B5	
D15	PC8	I/O	SDMMC1_D0	
E1	PE6 *	I/O	GPIO_Output	DIB5_CSA
E2	PE4	I/O	SPI4_NSS	DIB2_CSA
E3	VSS	Power		
E5	PB6	I/O	CEC	DVID_CEC
E6	VCAP4	Power		
E7	PD6	I/O	ETH_CLK	
E8	VSSUSB	Power		
E9	VSS	Power		
E10	PG1	I/O	LTDC_R6	
E11	PE12	I/O	GPIO_EXTI12	DVID_HPD
E12	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	SWDIO
E13	PA10	I/O	LTDC_B4	
E14	PC9	I/O	SDMMC1_D1	
E15	PC6 **	I/O	SDMMC1_D6	
F1	PG14 *	I/O	GPIO_Output	SYSMCU_CS
F2	PG12	I/O	LTDC_G1	

Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
F3	PG11	I/O	ETH_RMII_TX_EN	
F6	VSS	Power		
F7	VDD	Power		
F8	VDD33USB	Power		
F9	VDD	Power		
F10	VDD	Power		
F11	VCAP3	Power		
F12	PA8	I/O	LTDC_B6	
F13	PC7 **	I/O	SDMMC1_D7	
F14	PN1	I/O	XSPIM_P2_NCS1	
F15	VSS	Power		
G1	VLXSMPS	Power		
G2	VFBSMPS	Power		
G5	PG13	I/O	ETH_RMII_TXD0	
G6	VDD	Power		
G7	VDDLDO	Power		
G8	VSS	Power		
G9	VDDLDO	Power		
G10	VSS	Power		
G11	PA15(JTDI)	I/O	LTDC_R5	
G12	VSS	Power		
G13	PN3	I/O	XSPIM_P2_IO1	
G14	PN0	I/O	XSPIM_P2_DQS0	
G15	PN11	I/O	XSPIM_P2_IO7	
H1	VDDSMPS	Power		
H2	VSSSMPS	Power		
H4	PF7	I/O	UART7_TX	UART_TX/DOUT1
H5	PF6 *	I/O	GPIO_Output	DIB4_CSA
H6	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VDD	Power		
H11	VSS	Power		
H12	VDDXSPI2	Power		
H13	PN10	I/O	XSPIM_P2_IO6	
H14	PN9	I/O	XSPIM_P2_IO5	
H15	PN2	I/O	XSPIM_P2_IO0	
J1	VSS	Power		

Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
J2	PF9	I/O	LTDC_R0	
J3	PF10	I/O	LTDC_R1	
J4	NRST	Reset		
J5	VSSA	Power		
J6	VDDA	Power		
J7	VDDLDO	Power		
J8	VSS	Power		
J9	VDDLDO	Power		
J10	VSS	Power		
J11	VDDXSPI2	Power		
J12	VSS	Power		
J14	PN6	I/O	XSPIM_P2_CLK	
J15	VSS	Power		
K1	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
K2	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
K4	VREFM	Power		
K5	VREFP	Power		
K6	VDD	Power		
K7	VSS	Power		
K8	VDD	Power		
K9	VSS	Power		
K10	VDD	Power		
K11	VCAP2	Power		
K12	VDDXSPI2	Power		
K13	PN8	I/O	XSPIM_P2_IO4	
K14	PN4	I/O	XSPIM_P2_IO2	
K15	PN5	I/O	XSPIM_P2_IO3	
L1	PC1	I/O	ETH_MDC	
L2	PC2	I/O	SPI2_MISO	DIB1/DIB3_MISO
L3	PC3	I/O	SPI2_MOSI	DIB1/DIB3_MOSI
L4	PA2	I/O	ETH_MDIO	
L5	PC4	I/O	ETH_RMII_RXD0	
L6	VSS	Power		
L7	VCAP1	Power		
L8	VDDXSPI1	Power		
L9	VDD	Power		
L10	VDDXSPI1	Power		
L11	VSS	Power		
L12	PB12	I/O	LTDC_G5	

Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L14	PD14 *	I/O	GPIO_Output	USB_FS_SEL
M1	PA0	I/O	LTDC_G3	
M2	PA1	I/O	LTDC_G2	
M3	PA4	I/O	LTDC_R3	
M4	PA7	I/O	ETH_RMII CRS_DV	
M5	PF11	I/O	LTDC_B0	
M6	PE7	I/O	UART7_RX	UART_RX/DIN1
M7	VSS	Power		
M8	VDDXSPI1	Power		
M9	VSS	Power		
M10	VDDXSPI1	Power		
M11	PO3	I/O	XSPIM_P1_DQS1	
M12	PP10	I/O	XSPIM_P1_IO10	
M13	PB14	I/O	UART4_DE	RS485_TX_EN/DOUT2
M14	PB15	I/O	LTDC_G7	
M15	VSS	Power		
N1	PA3	I/O	LTDC_DE	
N2	PA5	I/O	LTDC_CLK	
N3	PC5	I/O	ETH_RMII_RXD1	
N4	PB2	I/O	LTDC_B2	
N5	PF15	I/O	SPI5_SCK	DIB4_SCK
N6	PP12	I/O	XSPIM_P1_IO12	
N7	PP14	I/O	XSPIM_P1_IO14	
N8	VSS	Power		
N9	PP2	I/O	XSPIM_P1_IO2	
N10	PP5	I/O	XSPIM_P1_IO5	
N11	PO2	I/O	XSPIM_P1_DQS0	
N12	PP1	I/O	XSPIM_P1_IO1	
N14	PB11	I/O	LTDC_G6	
N15	PB13	I/O	LTDC_G4	
P1	PA6	I/O	LTDC_B7	
P2	PB0 *	I/O	GPIO_Output	VMUX_EN
P3	PF13 *	I/O	GPIO_Output	VMUX_SEL
P4	PF14	I/O	LTDC_G0	
P6	PP11	I/O	XSPIM_P1_IO11	
P8	PP15	I/O	XSPIM_P1_IO15	
P9	PP3	I/O	XSPIM_P1_IO3	
P11	PP0	I/O	XSPIM_P1_IO0	
P12	PP7	I/O	XSPIM_P1_IO7	

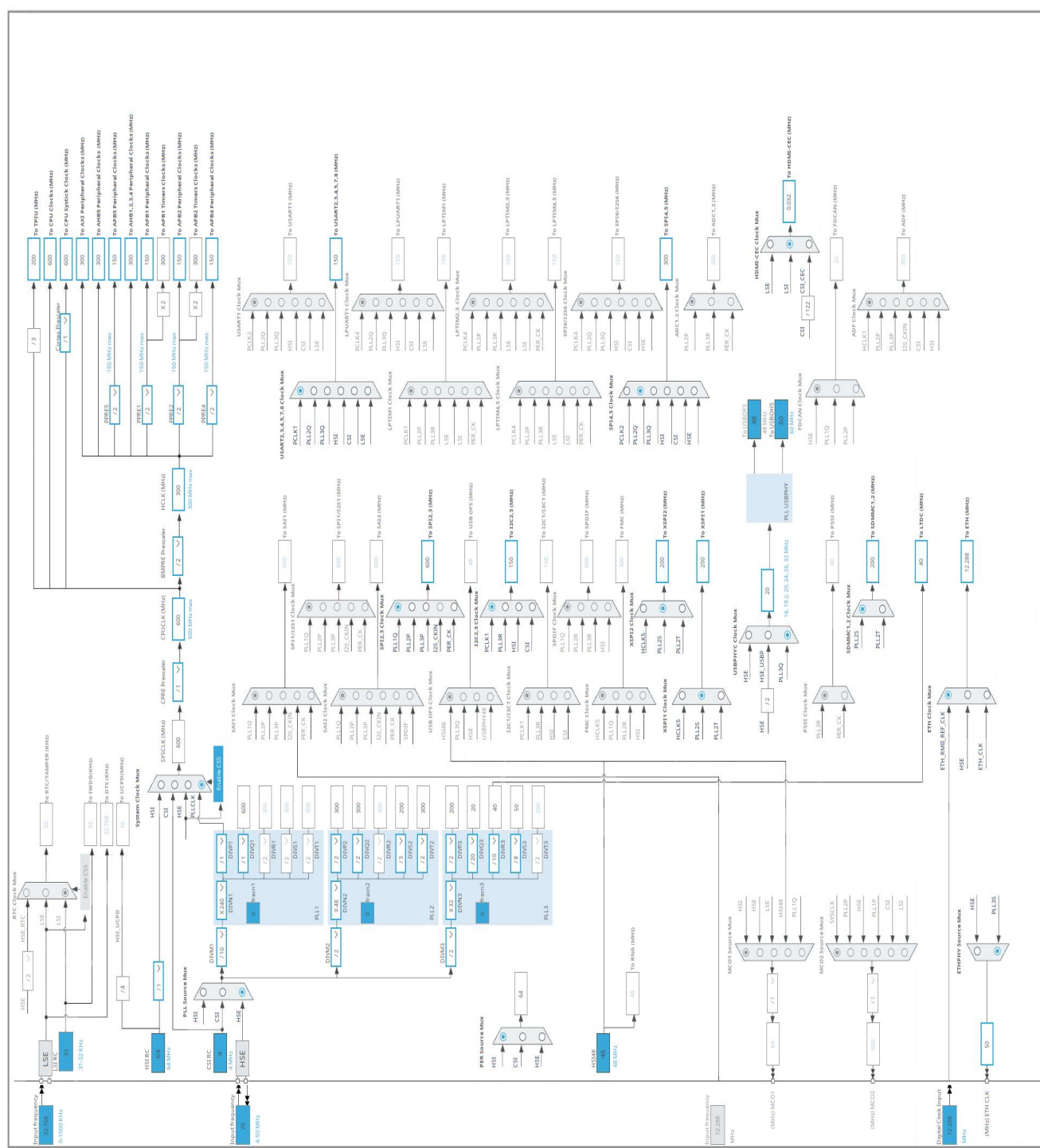
Pin Number TFBGA225 HEXA SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
P13	PP8	I/O	XSPIM_P1_IO8	
P15	PB10 **	I/O	ETH_MII_RX_ER	
R1	VSS	Power		
R2	PB1	I/O	GPIO_EXTI1	DIB1_IRQ
R4	PE8	I/O	GPIO_EXTI8	TFT_IRQ
R5	PE10 *	I/O	GPIO_Output	LED_nACT
R6	VSS	Power		
R7	PP13	I/O	XSPIM_P1_IO13	
R8	PP4	I/O	XSPIM_P1_IO4	
R9	VSS	Power		
R10	PO4	I/O	XSPIM_P1_CLK	
R11	PP6	I/O	XSPIM_P1_IO6	
R12	VSS	Power		
R13	PO0	I/O	XSPIM_P1_NCS1	
R14	PP9	I/O	XSPIM_P1_IO9	
R15	VSS	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32H7R7L8HxH



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H7R7/7S7
MCU	STM32H7R7L8HxH
Datasheet	DS000000 Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

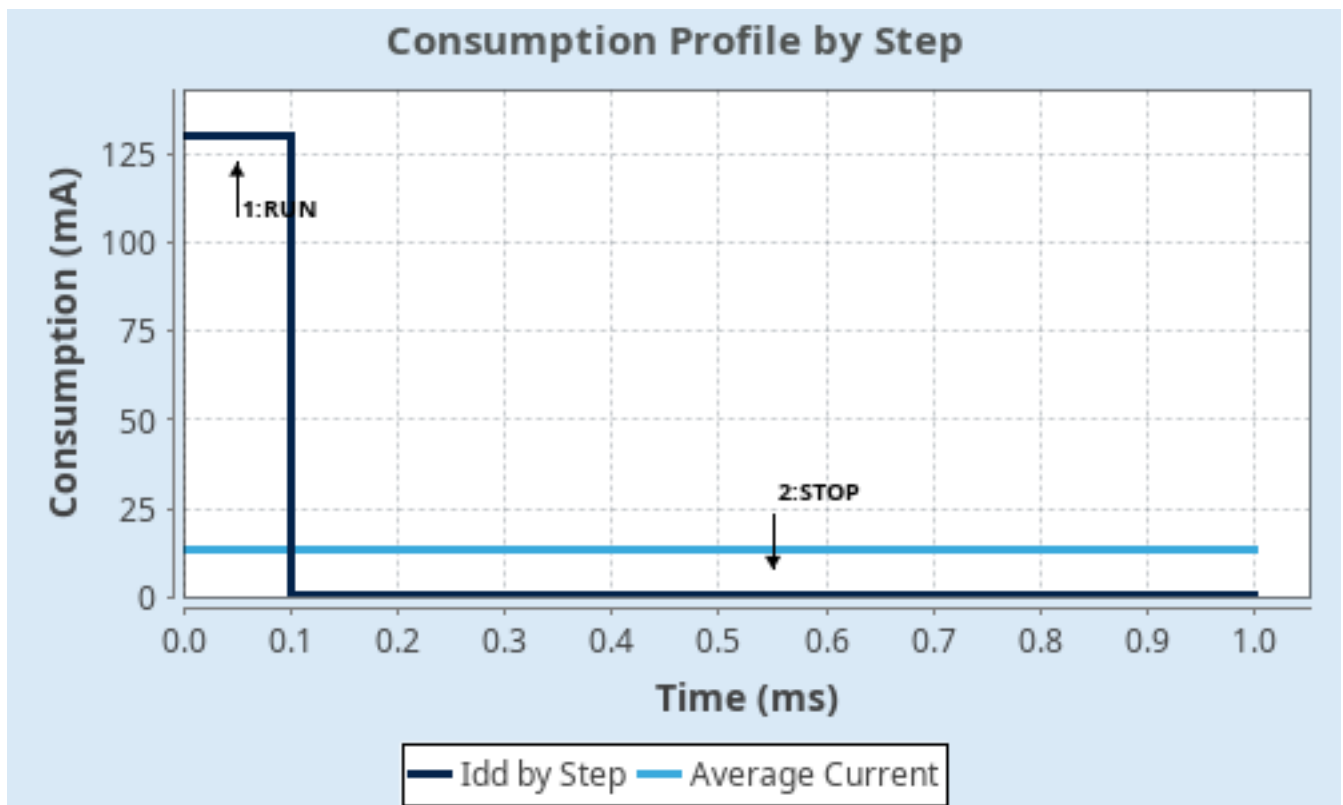
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS-HIGH	SVOS-LOW/SMPS
Fetch Type	AXISRAM3/FlashMode- ON/Cache/AlgoType- CoreMark	FLASH
CPU Frequency	600 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	130 mA	265 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1284.0	0.0
Ta Max	107.45	124.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	13.24 mA
Battery Life	1 day, 23 hours	Average DMIPS	128.40001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	H7R7L8H8
Project Folder	/home/denis/git-public/eez-bb3plus-cm5-firmware/ioc
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7RS V1.3.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x1000
Minimum Stack Size	0x1000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls Boot

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_FLASH_Init	FLASH
4	MX_XSPI1_Init	XSPI1
5	MX_XSPI2_Init	XSPI2
6	MX_SBS_Init	SBS
7	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

2.4. Advanced Settings - Generated Function Calls Appli

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA2D_Init	DMA2D
4	MX_I2C2_Init	I2C2
5	MX_JPEG_Init	JPEG
6	MX_LTDC_Init	LTDC
7	MX_SDMMC1_SD_Init	SDMMC1
8	MX_SPI2_Init	SPI2
9	MX_SPI4_Init	SPI4
10	MX_SPI5_Init	SPI5
11	MX_UART7_Init	UART7
12	MX_UART4_Init	UART4
13	MX_FLASH_Init	FLASH
14	MX_HDMI_CEC_Init	HDMI_CEC
15	MX_LWIP_Init	LWIP
16	MX_USB_DEVICE_Init	USB_DEVICE

2.5. Advanced Settings - Generated Function Calls ExtMemLoader

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7_APPLI

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode Disabled

Cortex Interface Settings:

CPU ICache Enabled *

CPU DCache Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers *

Cortex Memory Protection Unit Region 0 Settings:

MPU Region Enabled *

MPU Region Base Address 0x0 *

MPU Region Size 4GB *

MPU SubRegion Disable 0x87 *

MPU TEX field level level 0

MPU Access Permission ALL ACCESS NOT PERMITTED

MPU Instruction Access DISABLE *

MPU Shareability Permission ENABLE *

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region Enabled *

MPU Region Base Address 0x70000000 *

MPU Region Size 32MB *

MPU SubRegion Disable 0x0 *

MPU TEX field level level 1 *

MPU Access Permission ALL ACCESS PERMITTED *

MPU Instruction Access DISABLE *

MPU Shareability Permission DISABLE

MPU Cacheable Permission ENABLE *

MPU Bufferable Permission ENABLE *

Cortex Memory Protection Unit Region 2 Settings:

MPU Region Enabled *

MPU Region Base Address

	0x70000000 *
MPU Region Size	2MB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x90000000 *
MPU Region Size	32MB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x20000000 *
MPU Region Size	64KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x24000000 *
MPU Region Size	512KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *

MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x2406c000 *
MPU Region Size	16KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x24070000 *
MPU Region Size	8KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x25000000 *
MPU Region Size	16MB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.2. DEBUG

Debug: Serial Wire

3.2.1. Core(s) Settings:

Context(s): Boot
 Application

3.3. DMA2D

mode: Activated

3.3.1. Parameter Settings:

Basic Parameters:

Transfer Mode Memory to Memory

Color Mode ARGB8888

Output Offset	0
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Foreground layer Configuration:

DMA2D Input Color Mode ARGB8888

DMA2D ALPHA MODE	No modification of the alpha channel value
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Input Alpha	0
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Input Offset	0
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DMA2D ALPHA Inversion	Regular Alpha
DMA2D Red and Blue swap	Regular mode (RGB or ARGB)
DMA2D Chroma Sub-Sampling Mode	No chroma sub-sampling 4:4:4

3.4. ETH

Mode: RMII

3.4.1. Parameter Settings:

General : Ethernet Configuration:

Warning	The ETH can work only when RAM is pointing at 0x24000000
Note	PHY Driver must be configured from the LwIP 'Platform Settings' top right tab
Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
First Tx Descriptor Address	0x24020080 *
Rx Descriptor Length	4
First Rx Descriptor Address	0x24020000 *
Rx Buffers Address	0x24020100 *
Rx Buffers Length	1536

3.5. FLASH

mode: Enable

3.5.1. Parameter Settings:

BOR Level:

Activate	false
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User Configuration 1:

Activate	true *
IWDG_HW	enable *
NRST_STOP	no update
NRST_STDBY	no update
VDDIO_HSLV	no update
IWDG_FZ_STOP	no update
IWDG_FZ_SDBY	no update
XSPI1_HSLV	enable *
XSPI2_HSLV	enable *

User Configuration 2:

Activate	true
ECC_ON_SRAM	no update
I2C_NI3C	

DTCM size configuration:

Activate	false
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ITCM size configuration:

Activate	false
----------	-------

Write Protection:

Activate	false
----------	-------

HDP Protection:

Activate	false
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3.6. HDMI_CEC

mode: Activated

3.6.1. Parameter Settings:

Configuration parameters:

Signal Free Time	2.5, 4 or 6 nominal data bit periods
Rx tolerance	Standard tolerance
Signal Free Time option	SFT timer starts when Transmission Start Of Message is set by software
Listening mode	Receive all messages

Address parameters:

Logical address 0	Disable
Logical address 1	Disable
Logical address 2	Disable
Logical address 3	Disable
Logical address 4	Disable
Logical address 5	Disable
Logical address 6	Disable
Logical address 7	Disable
Logical address 8	Disable
Logical address 9	Disable
Logical address 10	Disable
Logical address 11	Disable
Logical address 12	Disable
Logical address 13	Disable
Logical address 14	Disable
Received data buffer name	cec_receive_buffer

Error handling:

Stop reception on bit rising error	Reception is stopped
Generate error bit on bit rising error	No error bit generation
Generate error bit on long bit period error	No error bit generation
Avoid error bit generation on error detection in broadcast	Error bit generation

3.7. I2C2

I2C: I2C

3.7.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x20C0EDFF *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.8. JPEG

mode: Activated

3.8.1. Parameter Settings:

Version:

JPEG version	jpeg1_v1_0
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JPEG Software options:

ENCODE	Enabled
DECODE	Enabled
RGB_FORMAT	JPEG_ARGB8888
JPEG_SWAP_RG	0

3.9. LTDC

Display Type: RGB888 (24 bits)

3.9.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	128 *
Horizontal Back Porch	88 *
Active Width	800 *
Horizontal Front Porch	40 *
HSync Width	127
Accumulated Horizontal Back Porch Width	215
Accumulated Active Width	1015
Total Width	1055

Synchronization for Height:

Vertical Synchronization Height	4
Vertical Back Porch	23 *
Active Height	600 *
Vertical Front Porch	1 *
VSynC Height	3
Accumulated Vertical Back Porch Height	26
Accumulated Active Height	626
Total Height	627

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	Active Low
Pixel Clock Polarity	Inverted Input *

Layer Default Color:

Red	0
Green	0
Blue	0

3.9.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha	255 *
-----------------	--------------

Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	255 *

Number of Layers:

Number of Layers	1 layer *
------------------	------------------

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	800 *
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	600 *

Pixel Parameters:

Layer 0 - Pixel Format	RGB888 *
------------------------	-----------------

Blending:

Layer 0 - Alpha constant for blending	255 *
Layer 0 - Blending Factor1	Alpha constant
Layer 0 - Blending Factor2	Alpha constant

Frame Buffer:

Layer 0 - Color Frame Buffer Start Address	0x90000000 *
Layer 0 - Color Frame Buffer Line Length (Image Width)	800 *
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	600 *

3.10. MEMORYMAP

mode: Activated

3.10.1. Core(s) Settings:

Context(s):	Boot
Initialized Context:	Boot
Power Domain:	

3.11. PWR

mode: USB REGULATOR

3.11.1. Parameter Settings:

USB_Regulator_Settings:

USB_REGEN	Disabled
USB_HS_REGEN	Disabled

3.12. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : BYPASS Clock Source

mode: Digital Clock Input (ETH_RMII_REF_CLK)

mode: Digital Clock Input (ETH_CLK)

3.12.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_DIRECT_SMPS_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	7 WS (8 CPU cycle)

PLL range Parameters:

PLL1 input frequency range	Between 2 and 4 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	High VCO range
PLL2 clock Output range	High VCO range
PLL3 clock Output range	Wide VCO range

3.13. SBS

mode: Activated

3.13.1. Parameter Settings:

Basic parameters:

IO HSLV FOR XSPIM1	ENABLE *
IO HSLV FOR XSPIM2	ENABLE *
IO HSLV FOR OTHER IO	

ENABLE *

3.14. SDMMC1

Mode: SD 4 bits Wide bus

3.14.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.15. SPI2

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.15.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	4 *
Baud Rate	150.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled
-----------------	----------

Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Output Hardware
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable

Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.16. SPI4

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.16.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	150.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled
-----------------	----------

Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Output Hardware
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.17. SPI5

Mode: Full-Duplex Master

3.17.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	128 *
Baud Rate	2.34375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled
-----------------	----------

Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.18. UART4

Mode: Asynchronous

mode: Hardware Flow Control (RS485)

3.18.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
----------------	----------------------

Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Polarity	High
Assertion Time	0
Deassertion Time	0

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.19. UART7

Mode: Asynchronous

3.19.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	9 Bits (including Parity) *
Parity	Even *
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable

Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.20. USB_OTG_HS

Internal HS Phy: Device_Only

3.20.1. Parameter Settings:

Speed	Device High Speed 480MBit/s
Enable internal IP DMA	Disabled
Physical interface	Internal HS Phy
Low power	Disabled
Link Power Management	Disabled
Use dedicated endpoint 1 interrupt	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

OTG PHY reference clock selection:

Ref Clock Selection	16 Mhz
---------------------	--------

3.21. XSPI1

Mode: Hexa SPI

Port: Port1 Hexa

Chip Select Override: NCS1 -- Port1 --

3.21.1. Parameter Settings:

Multiplexed:

Switching Duration Clock Number	1
---------------------------------	---

Generic:

Fifo Threshold	2 *
Memory Mode	Disable
Memory Type	16BITS AP Memory *
Memory Size	256 MBits *
Chip Select High Time Cycle	5
Free Running Clock	Disable
Clock Mode	Low

Wrap Size	32 Bytes *
Clock Prescaler	0
Sample Shifting	None
Chip Select Boundary	16 KBytes *
Maximum Transfer	0
Refresh Rate	400 *
Memory Select	NCS1

3.22. XSPI2

Mode: Octo SPI

Port: Port2 Octo

Chip Select Override: NCS1 -- Port2 --

3.22.1. Parameter Settings:

Generic:

Fifo Threshold	4 *
Memory Mode	Disable
Memory Type	Macronix *
Memory Size	256 MBits *
Chip Select High Time Cycle	2 *
Free Running Clock	Disable
Clock Mode	Low
Wrap Size	Not Supported
Clock Prescaler	0
Sample Shifting	None
Chip Select Boundary	Disabled
Maximum Transfer	0
Refresh Rate	0
Memory Select	NCS1

3.23. EXTMEM_MANAGER

mode: Activate External Memory Manager

3.23.1. Boot usecase:

Boot:

Select boot code generation	true *
-----------------------------	---------------

Selection of the boot system

Execute In Place

XIP:

select the memory

Memory 1

3.23.2. Memory 1:

Select driver:

Select the memory driver

EXTMEM_NOR_SFDP

Configuration:

Memory Instance

XSPI2 *

Number of memory data lines

EXTMEM_LINK_CONFIG_8LINES *

3.23.3. Memory 2:

Select driver:

Select the memory driver

EXTMEM_PSRAM *

Configuration:

Memory Instance

XSPI1

Number of memory data lines

EXTMEM_LINK_CONFIG_16LINES *

Memory Info:

Memory Size

HAL_XSPI_SIZE_256MB

Memory Maximun Frequency

200

Memory Command:

Read Register command

0x40 *

Write Register command

0xC0 *

Register Size

2

Register Dummy Cycles

4

Write Command

0xA0 *

Write Dummy Cycles

4

Read Command

0x20 *

Wrap Read Command

0x00 *

Read Dummy Cycles

4

Memory Configuration:

Number of Configuration Steps

1 *

Configuration 1:

Write Mask

0x40 *

Write Value	0x40 *
Register Address	0x08 *

3.24. LWIP

mode: Enabled

Advanced parameters are not listed except if modified by user.

3.24.1. General Settings:

LWIP Version:

LWIP Version (Version of LwIP supported by CubeMX ** CubeMX specific **)	2.2.1
--	-------

IPv4 - DHCP Options:

LWIP_DHCP (DHCP Module)	Enabled
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RTOS Dependency:

WITH_RTOS (Use FREERTOS ** CubeMX specific **)	Disabled
--	----------

RTOS_USE_NEWLIB_REENTRANT (No RTOS - 2)	Disabled
---	----------

Platform Settings:

PHY Driver	Choose/LAN8742
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Protocols Options:

LWIP_ICMP (ICMP Module Activation)	Enabled
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LWIP_IGMP (IGMP Module)	Disabled
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LWIP_DNS (DNS Module)	Disabled
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LWIP_UDP (UDP Module)	Enabled
-----------------------	---------

MEMP_NUM_UDP_PCB (Number of UDP Connections)	4
--	---

LWIP_TCP (TCP Module)	Enabled
-----------------------	---------

MEMP_NUM_TCP_PCB (Number of TCP Connections)	5
--	---

3.24.2. Key Options:

Network Interfaces Options:

LWIP_NETIF_STATUS_CALLBACK (Callback Function on Interface Status Changes)	Disabled
--	----------

LWIP_NETIF_EXT_STATUS_CALLBACK (Extended Callback Function for several netif)	Disabled
---	----------

LWIP_NETIF_LINK_CALLBACK (Callback Function on Interface Link Changes)	Enabled
--	---------

Infrastructure - OS Awareness Option:

NO_SYS (OS Awareness)	OS Not Used
-----------------------	-------------

Infrastructure - Timers Options:

LWIP_TIMERS (Use Support For sys_timeout)	Enabled
---	---------

Infrastructure - Core Locking and MPU Options:

SYS_LIGHTWEIGHT_PROT (Memory Functions Protection)	Disabled
--	----------

Infrastructure - Heap and Memory Pools Options:

MEM_SIZE (Heap Memory Size)	16384 *
ETH_RX_BUFFER_CNT	12

Infrastructure - Internal Memory Pool Sizes:

MEMP_NUM_PBUF (Number of Memory Pool struct Pbufs)	16
MEMP_NUM_RAW_PCB (Number of Raw Protocol Control Blocks)	4
MEMP_NUM_TCP_PCB_LISTEN (Number of Listening TCP Connections)	8
MEMP_NUM_TCP_SEG (Number of TCP Segments simultaneously queued)	16
MEMP_NUM_LOCALHOSTLIST (Number of Host Entries in the Local Host List)	1

Pbuf Options:

PBUF_POOL_SIZE (Number of Buffers in the Pbuf Pool)	16
PBUF_POOL_BUFSIZE (Size of each pbuf in the pbuf pool)	592

IPv4 - ARP Options:

LWIP_ARP (ARP Functionality)	Enabled
------------------------------	---------

Callback - TCP Options:

TCP_TTL (Number of Time-To-Live Used by TCP Packets)	255
TCP_WND (TCP Receive Window Maximum Size)	2144
TCP_QUEUE_OOSEQ (Allow Out-Of-Order Incoming Packets)	Enabled
LWIP_TCP_SACK_OUT (Allow Sending Selective Acknowledgements)	Disabled
TCP_MSS (Maximum Segment Size)	536
TCP_SND_BUF (TCP Sender Buffer Space)	1072
TCP_SND_QUEUELEN (Number of Packet Buffers Allowed for TCP Sender)	9

NETIF - Loopback Interface Options:

LWIP_NETIF_LOOPBACK (NETIF Loopback)	Disabled
--------------------------------------	----------

Thread Safe APIs - Socket Options:

LWIP_SOCKET (Socket API)	Disabled
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3.24.3. PPP:

PPP Options:

PPP_SUPPORT (PPP Module)	Disabled
--------------------------	----------

3.24.4. IPv6:

IPv6 Options:

LWIP_IPV6 (IPv6 Protocol)	Disabled
---------------------------	----------

3.24.5. HTTPD:

HTTPD Options:

LWIP_HTTPD (LwIP HTTPD Support ** CubeMX specific **) Disabled

3.24.6. SNMP:

SNMP Options:

LWIP_SNMP (LwIP SNMP Agent) Disabled

3.24.7. SNTP/SMTP:

SNTP Options:

LWIP_SNTP (LWIP SNTP Support ** CubeMX specific **) Disabled

SMTP Options:

LWIP_SMTP (LWIP SMTP Support ** CubeMX specific **) Disabled

3.24.8. MDNS/TFTP:

MDNS Options:

LWIP_MDNS (Multicast DNS Support ** CubeMX specific **) Disabled

TFTP Options:

LWIP_TFTP (TFTP Support ** CubeMX specific **) Disabled

3.24.9. Perf/Checks:

Sanity Checks:

LWIP_DISABLE_TCP_SANITY_CHECKS (TCP Sanity Checks) Disabled

LWIP_DISABLE_MEMP_SANITY_CHECKS (MEMP Sanity Checks) Disabled

Performance Options:

LWIP_PERF (Performace Testing for LwIP) Disabled

3.24.10. Statistics:

Debug - Statistics Options:

LWIP_STATS (Statistic Collection) Disabled

3.24.11. Checksum:

Infrastructure - Checksum Options:

CHECKSUM_BY_HARDWARE (Hardware Checksum ** CubeMX specific **)	Enabled
LWIP_CHECKSUM_CTRL_PER_NETIF (Generate/Check Checksum per Netif)	Disabled
CHECKSUM_GEN_IP (Generate Software Checksum for Outgoing IP Packets)	Disabled
CHECKSUM_GEN_UDP (Generate Software Checksum for Outgoing UDP Packets)	Disabled
CHECKSUM_GEN_TCP (Generate Software Checksum for Outgoing TCP Packets)	Disabled
CHECKSUM_GEN_ICMP (Generate Software Checksum for Outgoing ICMP Packets)	Enabled
CHECKSUM_GEN_ICMP6 (Generate Software Checksum for Outgoing ICMP6 Packets)	Disabled
CHECKSUM_CHECK_IP (Generate Software Checksum for Incoming IP Packets)	Disabled
CHECKSUM_CHECK_UDP (Generate Software Checksum for Incoming UDP Packets)	Disabled
CHECKSUM_CHECK_TCP (Generate Software Checksum for Incoming TCP Packets)	Disabled
CHECKSUM_CHECK_ICMP (Generate Software Checksum for Incoming ICMP Packets)	Enabled
CHECKSUM_CHECK_ICMP6 (Generate Software Checksum for Incoming ICMP6 Packets)	Disabled

3.24.12. Debug:

LwIP Main Debugging Options:

LWIP_DBG_MIN_LEVEL (Minimum Level)	All
------------------------------------	-----

3.24.13. Platform Settings:

Driver_PHY	LAN8742
------------	---------

3.25. USB_DEVICE

Class For HS IP: Communication Device Class (Virtual Port Com)

3.25.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
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USB CDC Tx Buffer Size

2048

3.25.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)

1155

LANGID_STRING (Language Identifier)

English(United States)

MANUFACTURER_STRING (Manufacturer Identifier)

STMicroelectronics

Device Descriptor HS:

PID (Product Identifier)

22336

PRODUCT_STRING (Product Identifier)

STM32 Virtual ComPort

CONFIGURATION_STRING (Configuration Identifier)

CDC Config

INTERFACE_STRING (Interface Identifier)

CDC Interface

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	SWCLK
	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	SWDIO
ETH	PD7	ETH_RMII_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	ETH_RMII_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG11	ETH_RMII_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG13	ETH_RMII_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC4	ETH_RMII_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA7	ETH_RMII_CRSDV	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC5	ETH_RMII_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
HDMI_CEC	PB6	CEC	Alternate Function Open Drain	No pull-up and no pull-down	Low	DVID_CEC
I2C2	PF0	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	SDA
	PF1	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	SCL
LTDC	PD3	LTDC_B1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB5	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG0	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB3(JTDO/TRACESWO)	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG2	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE11	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA11	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA9	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG1	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG12	LTDC_G1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA8	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA15(JTDI)	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PF9	LTDC_R0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF10	LTDC_R1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB12	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA0	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA4	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF11	LTDC_B0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB15	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA5	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB2	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB13	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA6	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF14	LTDC_G0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PC14-OSC32_IN(OSC32_IN)	RCC_OSC32_IN	n/a	n/a	n/a	
	PD6	ETH_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC1	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC8	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
SPI2	PA12	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB1/DIB3_SCK
	PB4(NJTRST)	SPI2_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB1_CSA
	PC2	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB1/DIB3_MISO
	PC3	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB1/DIB3_MOSI
SPI4	PE14	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB2/DIB5_MOSI
	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB2/DIB5_SCK
	PE13	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB2/DIB5_MISO
	PE4	SPI4_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB2_CSA
SPI5	PM13	SPI5_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB4_MOSI
	PM14	SPI5_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB4_MISO

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PF15	SPI5_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	DIB4_SCK
UART4	PD1	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_TX/DOUT3
	PD0	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_RX/DIN3
	PB14	UART4_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_TX_EN/DOUT2
UART7	PF7	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	UART_TX/DOUT1
	PE7	UART7_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	UART_RX/DIN1
USB_OTG_HS	PM6	USB_OTG_HS_DP	n/a	n/a	n/a	
	PM5	USB_OTG_HS_DM	n/a	n/a	n/a	
XSPI1	PO3	XSPIM_P1_DQS1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP10	XSPIM_P1_IO10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP12	XSPIM_P1_IO12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP14	XSPIM_P1_IO14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP2	XSPIM_P1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP5	XSPIM_P1_IO5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PO2	XSPIM_P1_DQS0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP1	XSPIM_P1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP11	XSPIM_P1_IO11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP15	XSPIM_P1_IO15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP3	XSPIM_P1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP0	XSPIM_P1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP7	XSPIM_P1_IO7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP8	XSPIM_P1_IO8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP13	XSPIM_P1_IO13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP4	XSPIM_P1_IO4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PO4	XSPIM_P1_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP6	XSPIM_P1_IO6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PO0	XSPIM_P1_NCS1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PP9	XSPIM_P1_IO9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
XSPI2	PN1	XSPIM_P2_NCS1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN3	XSPIM_P2_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN0	XSPIM_P2_DQS0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN11	XSPIM_P2_IO7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN10	XSPIM_P2_IO6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN9	XSPIM_P2_IO5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN2	XSPIM_P2_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
Single Mapped Signals	PN6	XSPIM_P2_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN8	XSPIM_P2_IO4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN4	XSPIM_P2_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN5	XSPIM_P2_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PM12	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB8	SDMMC1_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PM11	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PM9	USB_OTG_HS_ID	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB9	SDMMC1_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PM8	USB_OTG_HS_VBUS	n/a	n/a	n/a	
	PC6	SDMMC1_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC7	SDMMC1_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PB10	ETH_MII_RX_ER	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PE0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	SYSMCU_IRQ
	PF3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DIB3_IRQ
	PF2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DIB2_IRQ
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	WDG_OUT
	PM3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIB3_CSA
	PF4	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DIB4_IRQ
	PM2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DIN2
	PC15-OSC32_OUT(OSC32_OUT)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ETH_RESET
	PD5	GPIO_EXTI5	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DIB5_IRQ
	PM0	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	SD_DETECT
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_PWR
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIB5_CSA
	PE12	GPIO_EXTI12	External Interrupt Mode with Rising/Falling edge	No pull-up and no pull-down	n/a	DVID_HPD
	PG14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SYSMCU_CS
	PF6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIB4_CSA
	PD14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USB_FS_SEL

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	VMUX_EN
	PF13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	VMUX_SEL
	PB1	GPIO_EXTI1	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DIB1_IRQ
	PE8	GPIO_EXTI8	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	TFT_IRQ
	PE10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_nACT

4.2. GPDMA1

4.2.1. Core(s) Settings:

Context(s): Boot
Application
ExternalMemoryLoader

Initialized Context: ExternalMemoryLoader

Power Domain:

4.3. HPDMA1

4.3.1. Core(s) Settings:

Context(s): Boot
Application
ExternalMemoryLoader

Initialized Context: ExternalMemoryLoader

Power Domain:

4.4. LINKEDLIST

4.4.1. Core(s) Settings:

Context(s): Boot
Application
ExternalMemoryLoader

Initialized Context:

ExternalMemoryLoader

Power Domain:

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
EXTI line12 interrupt	true	0	0
I2C2 event interrupt	true	0	0
USB OTG HS interrupt	true	0	0
LTDC global interrupt	true	0	0
HDMI-CEC global interrupt	true	0	0
PVD and PVM interrupts through EXTI line	unused		
RCC global interrupt	unused		
FLASH interrupts	unused		
FPU global interrupt	unused		
EXTI line0 interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line3 interrupt	unused		
EXTI line4 interrupt	unused		
EXTI line5 interrupt	unused		
EXTI line8 interrupt	unused		
SPI2 global interrupt	unused		
SPI4 global interrupt	unused		
SPI5 global interrupt	unused		
I2C2 error interrupt	unused		
UART4 global interrupt	unused		
UART7 global interrupt	unused		
Ethernet global interrupt	unused		
LTDC Error global Interrupt	unused		
DMA2D global interrupt	unused		
JPEG global interrupt	unused		
SDMMC1 global interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI line12 interrupt	false	true	true
I2C2 event interrupt	false	true	true
USB OTG HS interrupt	false	true	true
LTDC global interrupt	false	true	true
HDMI-CEC global interrupt	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD and PVM interrupts through EXTI line	unused		
RCC global interrupt	unused		
FLASH interrupts	unused		
FPU global interrupt	unused		
XSPI1 global interrupt	unused		
XSPI2 global interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Category view Context Execution view Context Initialization view

Choose filters ...

... by Context Execution

☐ Boot ☐ Application ☐ ExternalMemoryLoader

... by Context Initialization

☐ Boot ☐ Application ☐ ExternalMemoryLoader ☒ None

Middleware

EXTMEM_MANAGER ✓

LWIP ✓

USB_DEVICE ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
CORTEX_M7_APPLI ✓			ETH ✓	DMA2D ✓			DEBUG ✓	PWR ✓	LINKEDLIST	SBS ✓
CORTEX_M7_BOOT			I2C2 ✓	HDMI_CEC ✓						
FLASH ✓			SDMMC1 ✓	JPEG ✓						
GPDMA1			SPI2 ✓	LTDC ✓						
GPIO ⚠			SPI4 ✓							
HPDMA1			SPI5 ✓							
NVIC_APPLI ✓			UART4 ✓							
NVIC_BOOT ✓			UART7 ✓							
RCC ✓			USB_HS ✓							
			XSPI1 ✓							
			XSPI2 ✓							

5.1.2. Without filters

Category view Context Execution view Context Initialization view

Choose filters ...

... by Context Execution

☐ Boot ☐ Application ☐ ExternalMemoryLoader

... by Context Initialization

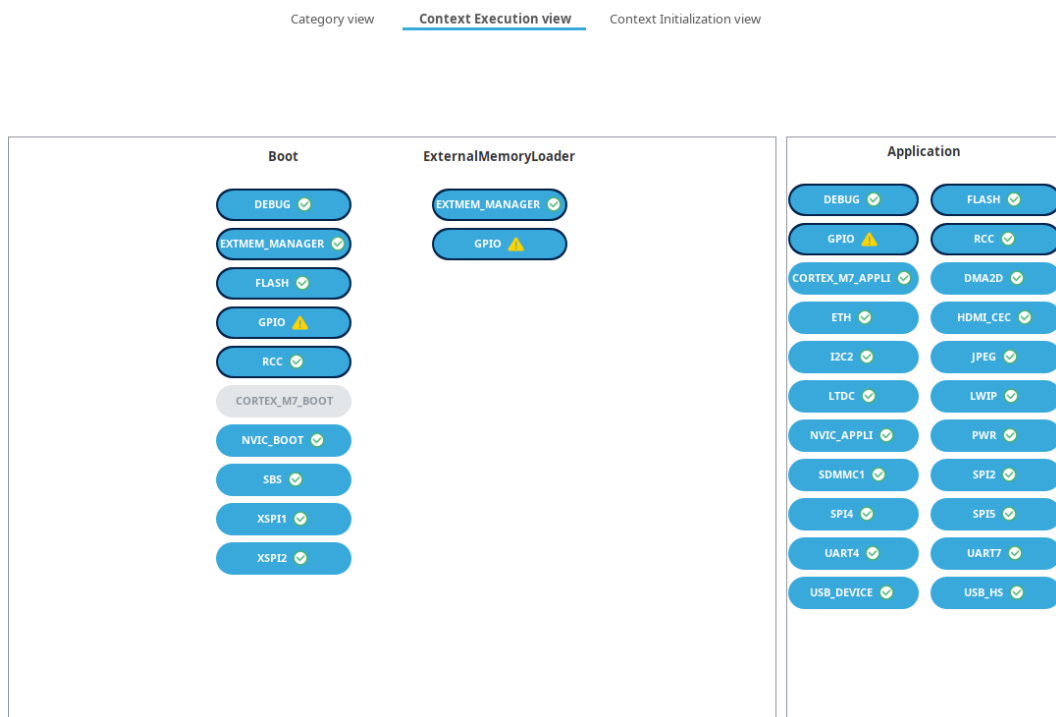
☐ Boot ☐ Application ☐ ExternalMemoryLoader ☒ None

Middleware

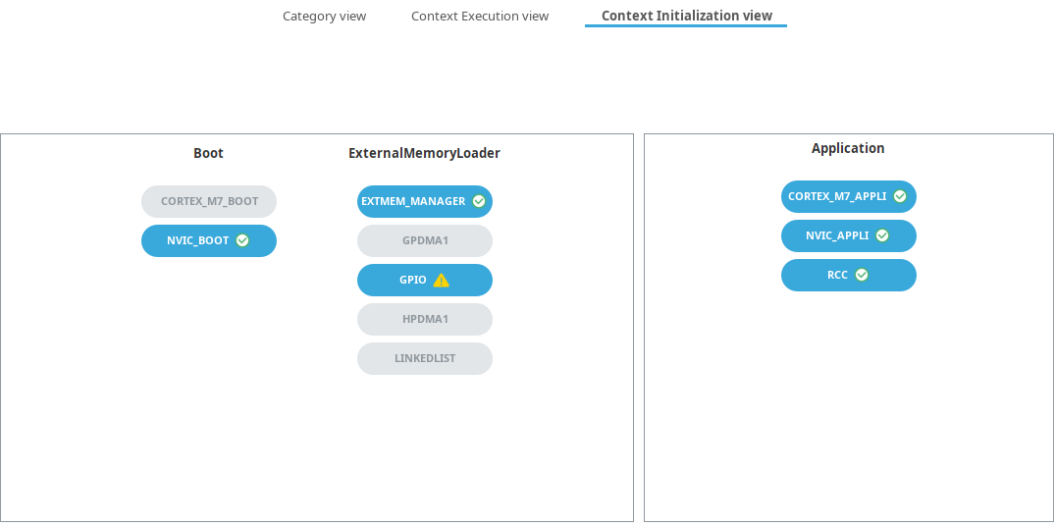
EXTMEM_MANAGER ✓ LWIP ✓ USB_DEVICE ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
CORTEX_M7_APPLI ✓			ETH ✓	DMA2D ✓			DEBUG ✓	PWR ✓	LINKEDLIST	SBS ✓
CORTEX_M7_BOOT			I2C2 ✓	HDMI_CEC ✓						
FLASH ✓			SDMMC1 ✓	JPEG ✓						
GPDMA1			SPI2 ✓	LTDC ✓						
GPIO ⚠			SPI4 ✓							
HPDMA1			SPI5 ✓							
NVIC_APPLI ✓			UART4 ✓							
NVIC_BOOT ✓			UART7 ✓							
RCC ✓			USB_HS ✓							
			XSPI1 ✓							
			XSPI2 ✓							

5.2. Context Execution view



5.3. Context Initialization view



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7rs-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
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Presentations	https://www.st.com/resource/en/product_presentation/stm32-graphics-solutions-detailed.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-

protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf

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secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf

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external-memory-manager-and-external-memory-loader-middleware-for-boot-flash-mcu-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
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